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Substituent effects on the electronic structures and fluorescent properties of salicylaldehyde acylhydrazone probes: A DFT/TDDFT study

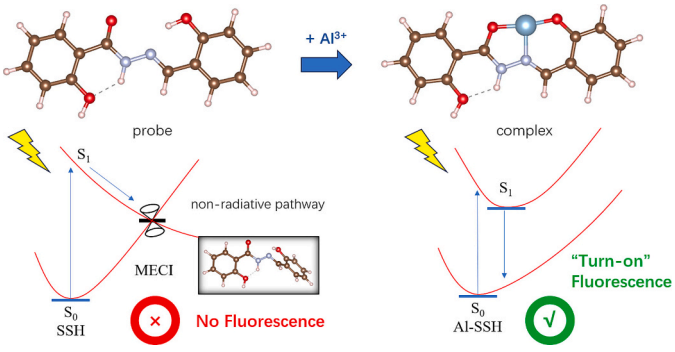
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HIGHLIGHTS

- DFT/TDDFT calculations reveal Al^{3+}-induced fluorescence turn-on mechanism.
- Free probes quenched via hydrazone rotation to conical intersection.
- ortho-OH group enhances planarity and intramolecular hydrogen bond.
- Naphthalene extension red-shifts absorption and emission.
- Metal coordination restricts hydrazone bond rotation, blocks the CI pathway.

GRAPHICAL ABSTRACT



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ABSTRACT

Salicylaldehyde acylhydrazone fluorescent probes exhibit excellent optical properties, making them valuable for fluorescence analysis and environmental monitoring. In this work, we investigate four representative probes (SBH, SSH, SNH, SNH-OH) to elucidate the substituent effects on their electronic structures and photophysical behavior. Our calculations reveal that all free probes are non-fluorescent due to an efficient non-radiative decay pathway mediated by rotation around the hydrazone bond, which leads to a conical intersection. Crucially, the introduction of an *ortho*-hydroxyl group enhances molecular planarity, while extending π-conjugation with a naphthalene ring significantly red shifts the absorption and emission. Upon binding with Al^{3+}, the hydrazone bond rotation is restricted, effectively blocking the nonradiative pathway and triggering "turn-on" fluorescence response. This work provides theoretical insights into how subtle structural modifications can precisely modulate the excited-state properties and thereby control the fluorescence of acylhydrazone-based sensors.

1. Introduction

Salicylaldehyde-derived Schiff bases are formed via the condensation of salicylaldehyde with primary amines or hydrazine derivatives [1,2]. They have excellent optical properties, along with the ability to form stable complexes with various metal ions, making them widely used as fluorescent probes for metal ion detection [3–5]. A variety of salicylaldehyde-based Schiff base sensors have been developed for the

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selective recognition of metal ions such as Cu^{2+}, Mg^{2+}, Zn^{2+}, and Al^{3+} [6–10]. For instance, a highly sensitive and selective dual ion sensing peptide-based Schiff base has been synthesized [11], and the probe exhibits "turn-on" detection of Zn^{2+} and Al^{3+}, with a detection limit as low as 1.25 nmol/L for Al^{3+}.

Among these Schiff bases, salicylaldehyde hydrazones can be further functionalized by introducing an acyl group, which adds an extra oxygen donor atom to the structure. This modification results in a more stable tridentate ligand known as salicylaldehyde acylhydrazone, and the structure is shown in Fig. 1. This structural motif has been extensively investigated in the literatures [12–16] as a good recognition unit for Al^{3+}. Notably, Wu et al. [17] have theoretically investigated the fluorescent properties of several salicylaldehyde acylhydrazone probes and designed two new near-infrared Al^{3+} fluorescent probes. In our previous work [18], we synthesized a fluorescent probe VN62 for detecting Al^{3+}. It shows high selectivity and sensitivity, excellent water solubility, and a wide range of pH response.

In recent years, fluorescence analysis has garnered significant attention for its speed, high sensitivity, good reproducibility, minimal sample consumption, and ease of operation. Usually, fluorescent probes are non-emissive in the absence of target analytes due to non-radiative decay pathways. However, upon binding to specific metal ions, the molecular rotation is restricted, leading to a significant enhancement in fluorescence emission, commonly referred as the "turn-on" effect [5]. Based on the photophysical processes of fluorescent probes, multiple photophysical mechanisms have been proposed to account for their sensing behavior [19], such as photoinduced electron transfer (PET) [20–22], excited state intramolecular proton transfer (ESIPT) [23–26], intramolecular charge transfer (ICT), including the special twisted ICT (TICT) state [27,28], and non-radiative decay pathways (e.g., C═N bond isomerization) leading to conical intersections (CI) [29–31]. Among these, the CI mechanism, which involves rotation around the hydrazone bond to access a degenerate S_0/S_1 region, is particularly relevant to the fluorescence quenching observed in many flexible Schiff base probes and serves as a primary focus of the present investigation.

The structures of probe molecules synthesized experimentally are often complex and adopt diverse conformations, with their fluorescent properties being highly sensitive to these structural variations. Despite the extensive experimental studies on Schiff base fluorescent probes, the understanding of their electronic structures, excited-state properties, and structure-activity relationships remains limited. This study aims to elucidate how substituents modulate the fluorescence emission of probe molecules by tuning molecular planarity, hydrogen bond strength, and the structure of the excited-state potential energy surface (PES), and to identify the probe structure that exhibits the strongest response to metal ions. To this end, we will perform theoretical calculations on the fluorescence mechanisms of four selected salicylaldehyde acylhydrazone derivatives and their complexes with metal ions. Such theoretical investigations can offer valuable insights into the molecular mechanisms underlying their fluorescence behavior, thereby supporting the rational design of highly sensitive and selective probes.

Calculation model and method.

Based on the structure of salicylaldehyde acylhydrazone, four probe molecules with different R substituents were designed: salicylaldehyde benzoyl hydrazone (SBH), salicylaldehyde salicyloyl hydrazone (SSH), salicylaldehyde naphthoyl hydrazone (SNH), and *ortho*-hydroxyl substituted SNH molecule (SNH-OH). The choice of these molecules was based on the following considerations: (1) Substituent effects. We aimed to explore the influence of different substituents (benzoyl, salicyloyl, naphthoyl) on the molecular conjugation, electronic structure, excited-state, and fluorescence properties. (2) Introduction of hydroxyl group. The addition of a hydroxyl group in SSH and SNH-OH was expected to form intramolecular hydrogen bonds, potentially affecting the molecular planarity and excited-state properties. (3) Conjugation extension. Replacing the benzene ring with a naphthalene was intended to expand the conjugated system, potentially enhancing the fluorescence properties. The optimized ground-state geometries of these compounds are shown in Fig. 2.

All calculations were performed using the ORCA 5.0 program package [32]. The ground-state geometries of the probe molecules and their Al^{3+} complexes were fully optimized using density functional theory (DFT). Subsequently, the first 5 singlet excited states were calculated using linear-response time-dependent DFT (LR-TDDFT). Given the significance of intramolecular hydrogen bond and potential charge-transfer character in the excited states, geometry optimizations for both ground state (S_0) and the first excited state (S_1) were carried out using the range-separated hybrid functional ωB97X-D3 [33], which includes empirical dispersion corrections. This functional is known for its reliable performance in describing non-covalent interactions and charge-transfer excitations in conjugated organic systems. The standard basis set 6-311G(d,p) was employed for all atoms. To account for solvation effects, all calculations were performed using the polarizable continuum model (PCM) to mimic an aqueous solution environment. Harmonic vibrational frequency calculations were performed for all ground-state optimized geometries. The absence of imaginary frequencies confirms that each structure is a true local minimum. For the S_1 minimum of SNH-OH, frequency analysis similarly shows no imaginary frequencies. The transition state (TS) of SNH-OH on the S_1 surface was confirmed by a single imaginary frequency.

To achieve better agreement with experimental absorption and emission energies, single-point TDDFT calculations using the M06 meta-GGA functional [34] were additionally performed on the ωB97X-D3 optimized geometries. Gaussian broadening with a width of 0.15 eV was applied in the absorption spectra to simulate the spectral broadening. The electron-hole distribution, reduced density gradient (RDG) [35,36], and atoms-in-molecules (AIM) [37] analyses were all performed using the Multiwfn toolkit [38]. The structures and molecular orbitals of the probes are displayed using VESTA [39] and VMD [40]. In addition, for the optimization of the minimum energy conical intersection (MECI) structures between the S_0 and S_1 states, the spin-flipped (SF) TDDFT method [41] was employed, as conventional single-reference TDDFT can fail to correctly describe the degenerate region near the CI.

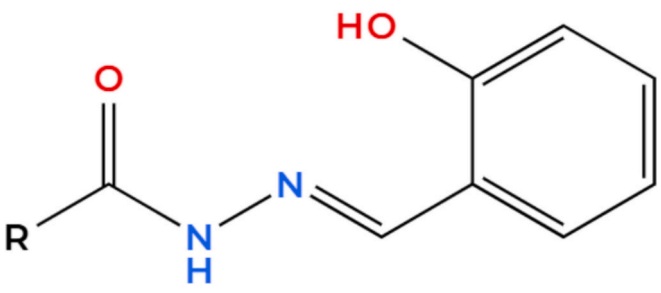


Fig. 1. Structure of salicylaldehyde acylhydrazone-based fluorescent probes.

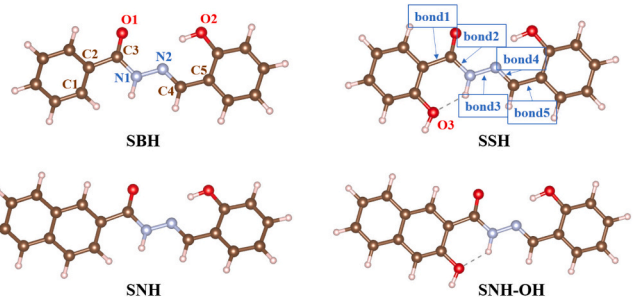


Fig. 2. Optimized ground-state geometries of the four probe molecules: SBH, SNH, SSH, and SNH-OH. Calculated at the ωB97X-D3/6-311G(d,p) level.

2. Results and discussion

2.1. Ground state properties

The salicylaldehyde acylhydrazone molecules exhibit significant flexibility, with multiple rotatable single bonds in the organic framework. Additionally, the rotation of double bonds such as amide and imine bonds can lead to cis-trans isomerism. To determine the most stable ground state structure of the molecules, we performed a flexible potential energy scan by rotating around the C2-C3 (bond1), C3-N1 (bond2), N1-N2 (bond3), and C4-C5 (bond5) bonds. Several local minima were identified, as shown in Fig. S1. Furthermore, considering the presence of various intramolecular hydrogen bonds in the probe molecules, new conformations can be formed when proton transfer (PT) occurs along the hydrogen bond direction. Based on the lowest energy rotational conformation, the O2H···N2 and N1H···O3 distances (only for SSH and SNH-OH) are scanned in the ground state. The energy curves of ground-state PT are shown in Fig. S2 and S3.

The structures shown in Fig. 2 are the final global minima (GM) in the ground state. The hydrogen bond lengths and relevant dihedral angles are listed in Table 1. The intramolecular O2H···N2 hydrogen bond is observed in all four probe molecules, and the salicylaldehyde acylhydrazone motif adopts a near-planar conformation, as indicated by the dihedral angle ∠C3N1-N2C4 being close to 180°. However, the angle between salicylaldehyde acylhydrazone and the substitutional parts differ significantly due to the varying R. Notably, SSH and SNH-OH adopt overall planar geometries. This planarity arises from the formation of an additional intramolecular N1H···O3 hydrogen bond. In contrast, the SBH and SNH molecules are non-planar, with the dihedral angle ∠C1C2-C3N1 being approximately 30°. To quantitatively characterize the intramolecular hydrogen bonding, we have additionally performed the RDG and AIM analyses (see Supporting Information Section 3, Fig. S4 and Table S1). The RDG isosurfaces clearly visualize the attractive interactions associated with the O2H···N2 and N1H···O3 hydrogen bonds. From the AIM analysis, the electron density at the bond critical point (ρ_b) for O2H···N2 follows the order SSH (0.0458) > SNH-OH (0.0454) > SNH (0.0449) > SBH (0.0403), indicating that SSH and SNH-OH exhibit stronger hydrogen bonding. This enhanced hydrogen-bonding strength is consistent with their shorter hydrogen bond lengths and more planar molecular geometries.

The molecular orbital energy levels and frontier molecular orbitals (HOMO and LUMO) of the salicylaldehyde acylhydrazone molecules were plotted. As shown in Fig. 3, the LUMO exhibits similar shapes and distributions, while the HOMO is related to the planarity of molecular structure. The HOMO energy levels of the four molecules are practically the same (around −8.2 eV), and the LUMO energy levels locate in a wide range (from −0.11 eV to 0.18 eV). Comparing between SBH and SNH, as well as SSH and SNH-OH, it was found that replacing the R group from a benzene ring to a naphthalene ring significantly reduces the LUMO energy, thereby decreasing the energy gap. Additionally, introducing a hydroxyl group at the C1 position has a minor effect on the LUMO energy level of SBH (from 0.18 eV to 0.12 eV), whereas for SNH, the LUMO energy level is markedly reduced (from −0.02 eV to −0.11 eV). In summary, whether by directly replacing the benzene ring with a naphthalene ring or by introducing a hydroxyl group, the conjugation of the molecular structure can be enhanced, leading to a reduced HOMO-LUMO gap.

Table 1
Selected ground-state geometric parameters of the probe molecules (SBH, SNH, SSH, SNH-OH). Listed are intramolecular hydrogen bond lengths, relevant dihedral angles, and planarity descriptors.

	SBH	SSH	SNH	SNH-OH
O2H···N2 (Å)	1.814	1.757	1.763	1.760
∠C3N1-N2C4 (°)	173.49	179.40	172.12	178.91
N1H···O3 (Å)	–	1.874	–	1.876
∠C1C2-C3N1 (°)	28.93	5.72	29.34	7.68

3. Absorption spectra

The absorption spectra with ωB97X-D3 functional of the acylhydrazone probe molecules were calculated and shown in Fig. S5. The excitation energies, oscillator strengths, and orbital transition contributions are listed in Table S2. According to the TDDFT results in Ref. [17], excitation energies obtained with the M06 functional agree better with experiment. Because meta-GGA functionals in ORCA have no analytic gradients, the ground- and excited-state geometries were optimized with ωB97X-D3, whereas the absorption and emission energies were recomputed with the more accurate M06 functional. Comparing the absorption spectra calculated with the two functionals shows that M06 and ωB97X-D3 give approximately the same shapes, with only a red-shift of the excitation energies.

The simulated absorption spectra, calculated using the M06 functional, are shown in Fig. 4. The absorption spectrum of SBH exhibits two peaks corresponding to S_1 and S_2 excitations at 312.3 nm and 276.8 nm, respectively, which are in good agreement with the experimental values (325 nm and 288 nm). The absorption profile of SSH is similar to that of SBH, with its S_1 transition slightly red shifted to 317.9 nm. Additionally, the oscillator strength of S_1 increases, while that of S_2 decreases. For SNH and SNH-OH, the maximum absorptions occur at 320.4 nm (S_1) and 325.5 nm (S_2), respectively, showing a more pronounced red shift compared to SBH, along with significantly enhanced absorption intensities.

The electron-hole distributions corresponding to the low energy absorption peaks of the series of probe molecules are shown in Fig. 5. TDDFT calculations at the M06/6-311G(d,p) level for the four salicylaldehyde acylhydrazone derivatives show same excited-state characters comparing with the ωB97X-D3 results. The S_1 states of SBH, SSH and SNH, together with the S_2 state of SNH-OH, correspond to a local excitation (LE1) localized on the salicylaldehyde acylhydrazone core. These transitions exhibit large oscillator strengths (0.638–0.893) and undergo a red-shift in the energy order: SBH (3.97 eV) > SSH (3.90 eV) > SNH (3.87 eV) > SNH-OH (S_2, 3.81 eV), which is consistent with the trend of their HOMO-LUMO energy gap variations. In contrast, the S_2 state of SNH and the S_1 state of SNH-OH arise from a second local excitation (LE2) confined to the naphthalene substituent. These LE2 states possess much smaller oscillator strengths (0.038 and 0.048, respectively). Introduction of a hydroxyl group on the naphthalene ring lowers the LE2 energy more strongly than the LE1 energy, so LE2 becomes the lowest-lying excited state (S_1) in SNH-OH, reversing the state ordering observed in SNH. Therefore, the nature of the excited-state PES can be tuned by simply modulating the substituent structure.

3.1. Excited states and conical intersections

According to Kasha's rule [42], fluorescence is typically emitted when a molecule relaxes from the first excited state back to the ground state; thus, the properties of the S_1 state determine the fluorescence emission characteristics of the system. Geometry optimizations of the S_1 states reveal a crucial difference among the probes: while SBH, SSH, and SNH lack any stable S_1 minimum and can directly access the S_0/S_1 CI region via rotation around hydrazone bond (bond4), the SNH-OH molecule exhibits a shallow local minimum on its S_1 surface. This minimum for SNH-OH corresponds to an excitation energy of 3.78 eV with an oscillator strength of 0.164. However, as will be confirmed by the later calculation of TS* optimization, the energy barrier separating this S_1 minimum from the CI region is very low (approximately 0.1 eV). Consequently, the non-radiative decay pathway through the CI is highly efficient and outcompetes fluorescence emission. Therefore, we conclude that all four free probe molecules (SBH, SSH, SNH, and SNH-OH) are effectively non-fluorescent due to facile access to the S_0/S_1

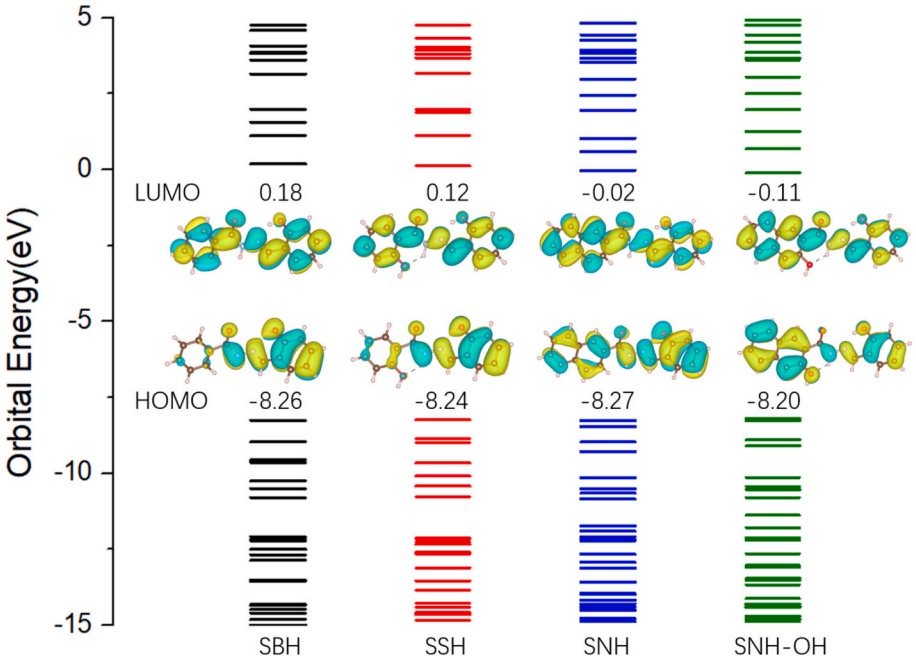


Fig. 3. Molecular orbitals energy levels and frontier orbitals (HOMO and LUMO) of the probe molecules.

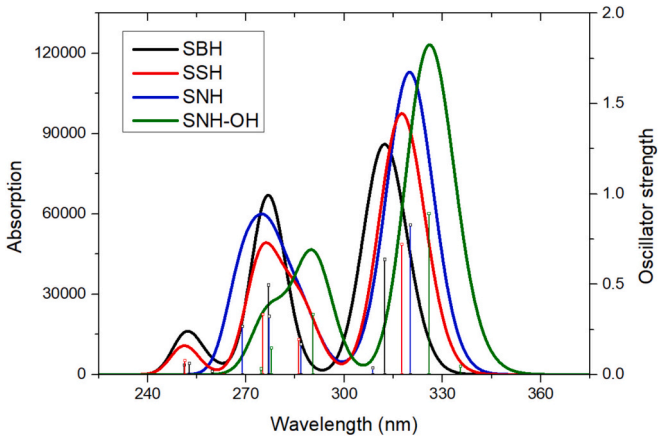


Fig. 4. Simulated absorption spectra of the free probe molecules, calculated with the M06/6-311G(d,p) functional on ωB97X-D3-optimized S_0 geometries.

CI. We optimized the MECI using the SF-TDDFT method, and the spin multiplicities of the S_0 and S_1 states were checked during the calculations, with values less than 0.6 for all cases. In the MECI structures, the dihedral angle ∠N1N2-C4C5 is twisted to approximately 90°. The MECI structures of the four probes have similar relative energies compared to the ground state, at around 3.4 eV. The relative energies of the critical points (Franck-Condon, S_1 minimum, and MECI) are summarized in Fig. 6. We have also verified these relative energies using a larger basis set def2-TZVPD, which yielded consistent results (see Supporting Information, Table S3), confirming the robustness of the PES topology.

To further analyze the photophysical processes of the probe molecules on the S_1 state and the influence of the dihedral angle ∠N1N2-C4C5 on the S_0/S_1 PES structure, the dihedral angle was scanned from −180° to 0°, and restricted optimizations of the molecules in the S_1 state were performed to plot the variation curve of excitation energy, as shown in Fig. 7. As the dihedral angle ∠N1N2-C4C5 increases, the energy of S_0 gradually rises, while that of S_1 gradually decreases (with SNH-OH showing an initial increase followed by a decrease). The excitation energy of the molecule diminishes to near 0 eV at a dihedral angle of approximately −120°. In the region between −120° and − 60°, the PES of S_0 and S_1 intersect. The energy curves of the four probes clearly demonstrate the existence of S_0/S_1 CI region and the characteristics of PES relations.

Based on the two-dimensional PES of SNH-OH excited states shown in Fig. S7, it can be seen that from the S_1 state to the CI region, the molecule undergoes a very small energy barrier TS*. To accurately determine the reaction barrier, TS optimization after Nudged Elastic Band (NEB) calculations were performed. The results indicate a negligible electronic barrier of 0.10 eV. Furthermore, Gibbs free energy analysis reveals that the TS* is energetically lower than the S_1 minimum ($\Delta G < 0$), indicating that the process is barrierless and spontaneous (k_{nr}), driven by entropic effects. According to the transition state theory [43], the rate constant for a barrierless process approaches the frequency factor limit, $k \approx k_B T/h$, which is approximately $6.21 \times 10^{12}\ s^{-1}$ at 298 K. Following the method we used in our previous work [44], we further estimated the radiative rate (k_r) using the Franck-Condon approximation, and compared it with the non-radiative decay rate via CI. The calculated k_r is $1.01 \times 10^{8}\ s^{-1}$. This large discrepancy ($k_{nr} \gg k_r$) provides a quantitative basis for the fluorescence quenching for SNH-OH. However, when the probe molecule binds to the target metal ion, the rotation of hydrazone bond is hindered, preventing the formation of CI. As a result, the S_1 emission state becomes stable, and fluorescence will be activated.

3.2. Interaction between probes and metal ions

As fluorescent probes, the acylhydrazone molecules can function as coordination ligands and bind to a variety of metal ions. To assess their binding affinity, we calculated the thermodynamic parameters for the following coordination reactions:

$M^{2+} + L^- = ML^+$; $M^{3+} + L^- = ML^{2+}$.

where L^- is the deprotonated probe ligand, M represents the metal ions (including Mg, Al, Cu, Zn, and Fe), and ML^+/ML^{2+} is the resulting metal complex. The corresponding reaction Gibbs free energy ΔG are given in Table 2. Six metal ions were selected to evaluate the strength of complex formation. For Fe^{2+} and Fe^{3+}, the calculations were performed using their low-energy high-spin electronic configurations. The results show that when $M^{n+} = Al^{3+}/Fe^{3+}$, the coordination reactions exhibit

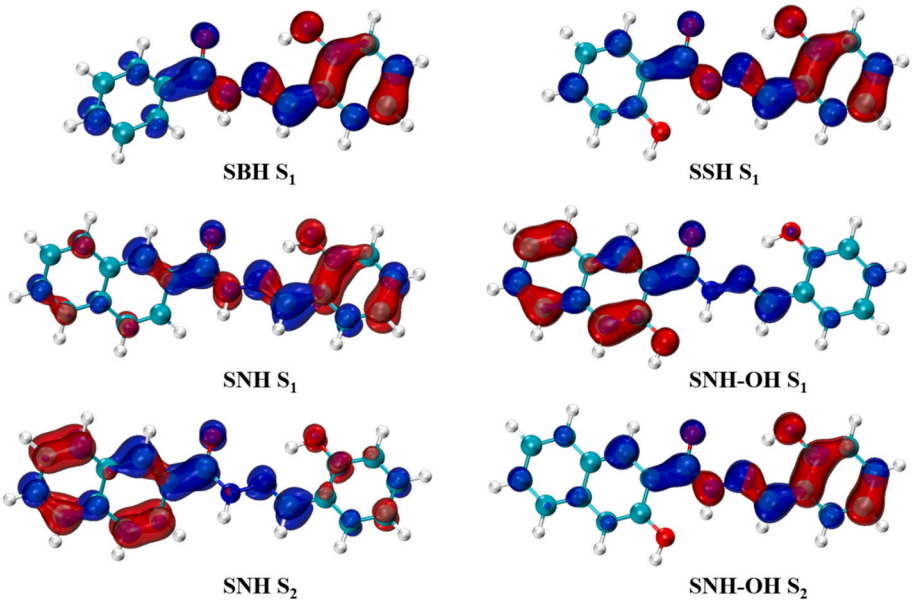


Fig. 5. Electron–hole distribution maps for the primary absorption transitions of SBH, SNH, SSH, and SNH-OH (using M06 functional). Blue and red isosurfaces represent electron and hole densities, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

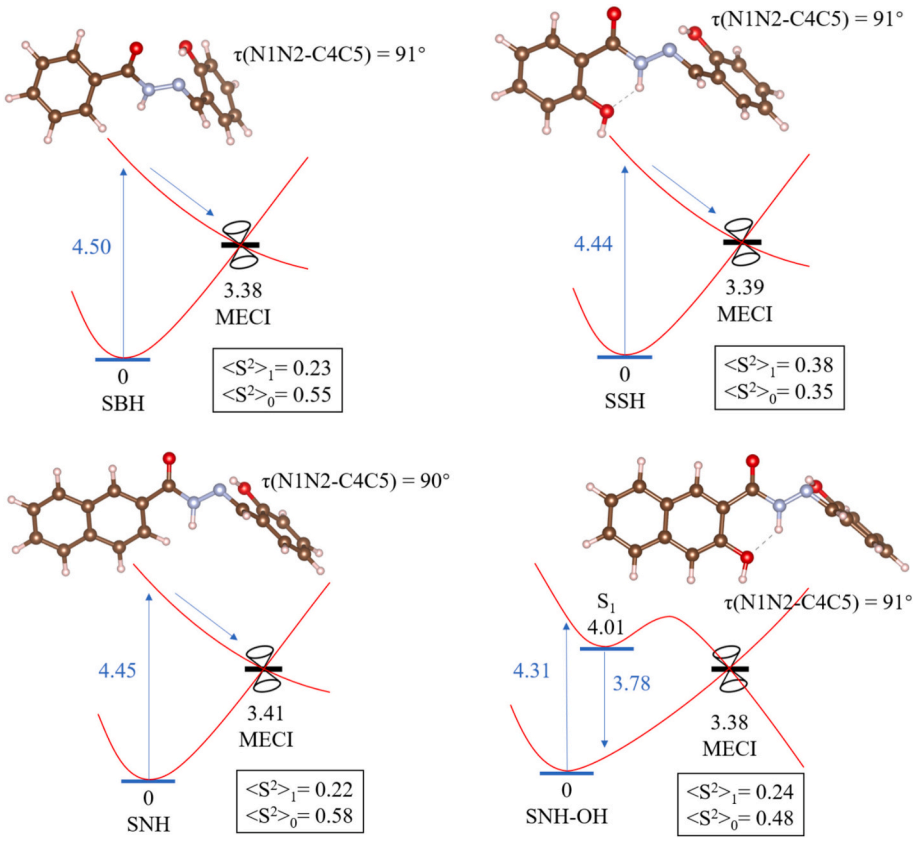


Fig. 6. Schematic potential energy diagram along the S_0 and S_1 states for the probe series, showing the relative energies of the Franck–Condon (FC) point, the S_1 minimum (SNH-OH), and the MECI between S_0/S_1. Nonradiative decay via C═N bond rotation leads to rapid internal conversion for SBH, SNH, and SSH.

significantly more negative ΔG values (lower than −350 kcal/mol), indicating thermodynamically spontaneous and strong binding. In contrast, binding with other metal ions is less favorable. The binding free energies for the four ligands are similar for a given metal ion, with SSH and SNH-OH showing slightly stronger binding due to their enhanced planarity and additional hydrogen bonding.

Considering that L may form a 2:1 complex with metal ions, we also calculated the ΔG for the reaction: $ML^{n-1} + L^{-} = ML_2^{n-2}$. Since the binding energies of the four ligands to the same metal ion are similar, we only performed calculations for L = SBH with different metal ions.

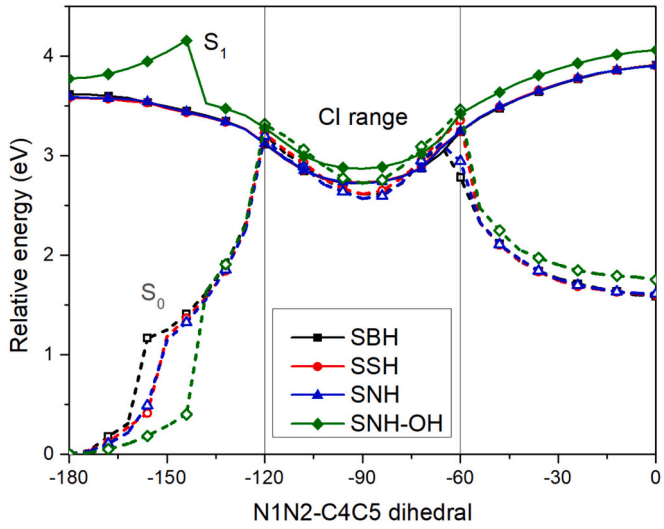


Fig. 7. Ground- (S_0, dashed line) and first excited-state (S_1, solid line) potential energy curves as a function of the dihedral ∠N1N2-C4C5 for the probe molecules, calculated at the ωB97X-D3/6-311G(d,p) level.

Table 2
Calculated Gibbs free energies of binding (ΔG, kcal/mol) for the formed between the probe molecules and metal ions, evaluated at 298 K using the ωB97X-D3/6-311G(d,p)//PCM approach.

ΔG (kcal/mol)	SBH	SSH	SNH	SNH-OH
Mg^{2+}	−117.05	−119.60	−117.22	−118.10
Al^{3+}	−358.58	−363.07	−359.54	−361.34
Cu^{2+}	−118.59	−121.21	−118.53	−119.77
Zn^{2+}	−93.43	−95.77	−93.38	−94.33
Fe^{2+}	−172.44	−176.09	−173.27	−174.61
Fe^{3+}	−406.15	−409.94	−406.64	−408.24

Additionally, to account for the influence of hydration on coordination in actual reactions, all ions were modeled in a six-coordinated structure. The binding free energy for different metal ions was calculated according to the following equation: $M(H_2O)_6^{n+} + L^- = [M(H_2O)_3L]^{n-1} + 3H_2O$. The energies of the latter two coordination reactions are listed in Table S5. It can be seen that Al^{3+}/Fe^{3+} still exhibit the strongest binding to ligand L, which is related to the high charge density of these two ions. Since Fe^{3+} itself is colored, and the fluorescence can be quenched due to d–d transitions, the fluorescence emission of the metal-bound molecules will be further calculated using the Al^{3+} complex as the model fluorescent complex.

3.3. *Fluorescent properties of the $[AlL]^{2+}$ complex*

Upon interaction with Al^{3+}, the probe molecule forms a coordination complex, triggering a "turn-on" fluorescence response. Al^{3+} binds strongly to the ligand (L), and the geometry of the resulting $[AlL]^{2+}$ complex has been optimized at the ground state level. Based on the optimized structures, TDDFT calculations were performed to investigate both the absorption and emission properties of the complex. The excitation details for $[AlL]^{2+}$ are summarized in Table S6, while the absorption spectra are displayed in Fig. S8. The calculated and experimental emission wavelengths are listed in Table 3. The calculated emission wavelengths for the $[AlL]^{2+}$ complex show good agreement with experimental observations: SBH emits at 418.1 nm (Exp. 452 nm), and SSH at 420.6 nm (Exp. 450 nm). In contrast, SNH and SNH-OH exhibit significantly red-shifted emissions at 480.4 nm and 559.8 nm, respectively, attributed to the electron-donating effects of their substituents. This trend in emission energy follows the order: SBH > SSH > SNH > SNH-OH, consistent with the corresponding absorption of the free probes.

Table 3
Calculated and experimental emission wavelengths (λ_{em}, nm) for the $[AlL]^{2+}$ complexes, calculated at the M06/6-311G(d,p) level based on ωB97X-D3-optimized S_1 geometries. (Experimental emission wavelengths for SNH and SNH-OH are not available as these complexes have not been reported yet.)

$[AlL]^{2+}$	SBH	SSH	SNH	SNH-OH
Calc. λ_{em} (nm)	418.1	420.6	480.4	559.8
Osc. strength	0.589	1.050	0.845	0.071
Exp. λ_{em} (nm)	452 [45]	450 [46]	–	–

To further elucidate the nature of the excited state, excitation analysis was conducted for the $[AlL]^{2+}$ complex. The electron-hole distribution of $[AlL]^{2+}$ emission state is illustrated in Fig. 8 (M06) and Fig. S10 (ωB97X-D3). Both functionals yield consistent results, confirming that the excitation is predominantly ligand-based. Notably, the oscillator strengths of the $[AlL]^{2+}$ complexes formed by SBH, SSH, and SNH are significantly large, indicating enhanced radiative transitions. However, the Al^{3+} + SNH-OH complex exhibits a small oscillator strength. This is consistent with the minimal spatial overlap between the electron and hole distributions in its S_1 state (Fig. 8), which arises from the reversed state ordering (LE2 being the S_1 state) in the free probe induced by the hydroxyl substituent.

Therefore, coordination with Al^{3+} induces strong fluorescence in the complexes, primarily driven by the ligand moiety. The emission characteristics are thus largely determined by the electronic structure and substitution pattern of the ligand, with the nature of the excited state S_1 playing a crucial role in dictating the photophysical behavior of the system.

4. Conclusions

In summary, we have designed and theoretically investigated four salicylaldehyde acylhydrazone fluorescent probes (SBH, SNH, SSH, SNH-OH) to elucidate the impact of substituent on their photophysical behavior. Our DFT/TD-DFT calculations reveal that:

(1) The introduction of an *ortho*-hydroxyl group on the acyl substituent (as in SSH and SNH-OH) facilitates the formation of an additional N1H···O3 hydrogen bond. This interaction significantly enhances molecular planarity, strengthens intramolecular hydrogen bonding.
(2) Extending the π-conjugation by replacing the benzene ring with a naphthalene unit (as in SNH and SNH-OH) effectively lowers the LUMO energy level, resulting in a pronounced red-shift of both absorption and emission spectra.
(3) The fluorescence of the free ligands is universally quenched via a non-radiative decay channel involving rotation around the hydrazone bond, which leads to CI nonradiative pathway. For SNH-OH, although a shallow local minimum exists on its S_1 PES, the free energy barrier from this minimum to the CI is negligible (ΔG < 0), rendering the non-radiative decay a barrierless and highly efficient process.
(4) Coordination with Al^{3+} rigidifies the molecular structure by restricting hydrazone bond rotation, thereby blocking the CI-mediated nonradiative pathway and inducing a dramatic "turn-on" of enhanced fluorescence. Notably, the nature of the S_1 state is crucial: the reversed state ordering in SNH-OH (where a low-oscillator-strength LE2 state becomes S_1) results in a weaker emission for its Al^{3+} complex compared to SBH, SSH, and SNH.

Overall, the probe SSH has the optimal structure, as it achieves an ideal balance: the *ortho*-hydroxyl group provides sufficient rigidification and stronger Al^{3+} binding without introducing low-lying, problematic excited states. This results in the brightest emission among the series.

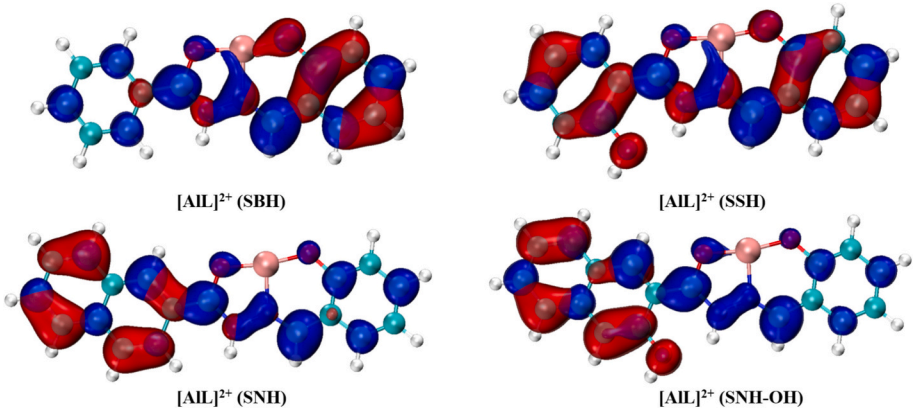


Fig. 8. Electron-hole distribution for the emission S_1-S_0 transition of the $[AlL]^{2+}$ complex, calculated at the M06/6-311G(d,p) level. The emission is predominantly ligand-centered character.

Our study demonstrates that the photophysical property of the probe is exquisitely sensitive to their molecular structure. By simply altering the substituent on the acylhydrazone framework, one can precisely engineer the excited-state PES, including tuning the excited-states order, controlling the presence or absence of stable minima on the S_1 surface, and modulating the accessibility of CI pathway. In conclusion, strategic substituent selection can serve as a powerful tool to tailor the fluorescence properties of salicylaldehyde acylhydrazone probes to achieve desired sensing performance.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used DeepSeek in order to improve the language and readability of the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.saa.2026.127974.

Data availability

The data supporting this study are available in Zenodo. https://doi.org/10.5281/zenodo.18205913

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